

Problem 2: Justification

n) Solution

$$|\psi\rangle_{AB} = \sum_i |i\rangle_A \otimes |i\rangle_B$$

$$S = \text{Tr}_B(|\psi\rangle_{AB} \langle\psi|_{AB})$$

For - The First

$$\frac{1}{2} I = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1|$$

$$|\psi\rangle_{AB} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)_{AB}$$

$$S = \text{Tr}_B(|\psi\rangle_{AB} \langle\psi|_{AB})$$

$$= \frac{1}{2} [\langle 0| (|\psi\rangle_{AB} \langle\psi|_{AB}) |0\rangle + \langle 1| (|\psi\rangle_{AB} \langle\psi|_{AB}) |1\rangle]$$

$$= \frac{1}{2} [|0\rangle\langle 0| + |1\rangle\langle 1|]$$

$$= \frac{1}{2} [|0\rangle\langle 0| + |1\rangle\langle 1|] = \frac{1}{2} I$$

For - Second

$$\frac{1}{2} I = \frac{1}{2} |1\rangle\langle 1| + \frac{1}{2} |-1\rangle\langle -1|$$

$$|\psi\rangle_{AB} = \frac{1}{\sqrt{2}} |1\rangle_A |0\rangle_B + \frac{1}{\sqrt{2}} |-1\rangle_A |1\rangle_B = \frac{|1\rangle_A |0\rangle_B + |-1\rangle_A |1\rangle_B}{\sqrt{2}}$$

$$S = \text{Tr}_B(|\psi\rangle_{AB} \langle\psi|_{AB})$$

$$= \frac{1}{2} [\langle 0| (|\psi\rangle_{AB} \langle\psi|_{AB}) |0\rangle + \langle 1| (|\psi\rangle_{AB} \langle\psi|_{AB}) |1\rangle]$$

$$= \frac{1}{2} [|1\rangle\langle 1| + |-1\rangle\langle -1|]$$

$$= \frac{1}{2} [|1\rangle\langle 1| + |-1\rangle\langle -1|] = \frac{1}{2} I$$

For - True Answer

$$\langle \psi | A_B = \frac{1}{2} (107127 + 117127 + 117137 + 1-7147) A_B$$

$$S = \text{Tr}_B (|\psi\rangle_{AB} \langle\psi|_{AB})$$

$$= \frac{1}{2} [211 (1\psi_{AB} \langle\psi|_{AB}) 127 + 221 (1\psi_{AB} \langle\psi|_{AB}) 127 + 231 (1\psi_{AB} \langle\psi|_{AB}) 137 + 44 (1\psi_{AB} \langle\psi|_{AB}) 147]$$

$$= \frac{1}{2} [107201 + 117221 + 117221 + 1-7221] = \frac{1}{2} I$$

2) Solution

$$S_A = \begin{pmatrix} p & 0 \\ 0 & 2-p \end{pmatrix} \text{ where } 0 \leq p \leq 1$$

$$|\psi\rangle_{AB} = \sqrt{p} |0\rangle_A \otimes |0\rangle_B + \sqrt{2-p} |1\rangle_A \otimes |1\rangle_B$$

The reduced state

$$S_A = \text{Tr}_B (|\psi\rangle_{AB} \langle\psi|_{AB})$$

$$= p (201 (1\psi_{AB} \langle\psi|_{AB}) 107 + (2-p) (221 (1\psi_{AB} \langle\psi|_{AB}) 127)$$

$$= p (107201) + (2-p) (127221)$$

$$= p \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + (2-p) \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} p & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 2-p \end{bmatrix} = \begin{pmatrix} p & 0 \\ 0 & 2-p \end{pmatrix}$$

$$= \begin{bmatrix} p & 0 \\ 0 & 2-p \end{bmatrix}$$

Therefore, unitization

$$\begin{aligned} |\psi\rangle_{AB} &= \sqrt{p} |\psi\rangle_A \otimes |\psi\rangle_B + \sqrt{1-p} |\psi\rangle_A \otimes |\psi\rangle_B \\ &= \sqrt{p} |\psi\rangle_A \otimes |\psi\rangle_B + \sqrt{1-p} |\psi\rangle_A \otimes |\psi\rangle_B \\ &\quad \underline{\underline{\hspace{10em}}}\end{aligned}$$

3) Solution

Density operator

$$\rho_A = \sum_i \lambda_i |\psi_i\rangle_A \langle \psi_i|, \quad \lambda_i \geq 0$$

- Two unitization $|\psi\rangle_{AB}$ and $|\phi\rangle_{AB}$ in $\mathcal{H}_A \otimes \mathcal{H}_B$

$$\rho_A = \text{Tr}_B (|\psi\rangle_{AB} \langle \psi|_{AB})$$

$$\rho_A = \text{Tr}_B (|\phi\rangle_{AB} \langle \phi|_{AB})$$

One orthonormal vector

$$|\psi\rangle_{AB} = \sum_i \sqrt{\lambda_i} |\psi_i\rangle_A \otimes |\psi_i\rangle_B$$

$$|\phi\rangle_{AB} = \sum_i \sqrt{\lambda_i} |\psi_i\rangle_A \otimes |\psi_i\rangle_B$$

Define the linear operator U_B on the span of $\{|\psi_i\rangle_B\}$

$$U_B |\psi_i\rangle_B = |\psi_i\rangle_B \text{ for each } i$$

Now apply $(I_A \otimes U_B)$ on $|\psi\rangle_{AB}$

$$(I_A \otimes U_B) |\psi\rangle_{AB} = \sum_i \sqrt{\lambda_i} |\psi_i\rangle_A \otimes U_B |\psi_i\rangle_B$$

$$(\sum_A \otimes U_B) |\psi\rangle_{AB} = \sum_i \bar{A}_i |\psi\rangle_A \otimes |U_i\rangle_B \\ = |\psi\rangle_{AB}$$

Hence,

$$|\psi\rangle_{AB} = (\sum_A \otimes U_B) |\psi\rangle_{AB}$$

Uhlmann's theorem

$$F(\rho, \sigma) = \max_{|\psi\rangle, |\phi\rangle} |\langle \psi | \phi \rangle|$$

If $|\psi_1\rangle$ and $|\psi_2\rangle$ are two purification of the same state ρ

$$|\psi_1\rangle = (I \otimes U) |\psi_2\rangle$$

Purification of each state

$$|\bar{\psi}\rangle \text{ of } \rho$$

$$|\bar{\phi}\rangle \text{ of } \sigma$$

$$|\psi\rangle = (I \otimes U(\psi)) |\bar{\psi}\rangle$$

$$|\phi\rangle = (I \otimes U(\phi)) |\bar{\phi}\rangle$$

$$\langle \psi | \phi \rangle = \langle \bar{\psi} | (I \otimes U^\dagger(\psi)) (I \otimes U(\phi)) |\bar{\phi}\rangle, \quad U = U^\dagger(\psi) U(\phi)$$

$$\langle \psi | \phi \rangle = \langle \bar{\psi} | (I \otimes U) |\bar{\phi}\rangle, \quad U = U^\dagger(\psi) U(\phi)$$

Maximization reduce to single unitary

$$F(\rho, \sigma) = \max_U |\langle \bar{\psi} | (I \otimes U) |\bar{\phi}\rangle|$$

Problem 2: Properties of fidelity

a) Solution

; Show that: $F(\rho, \sigma) = F(\rho, \sigma)$

$$F(\rho, \sigma) = \text{Tr} \left[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right]$$

Spectral decomposition

$$\sqrt{\rho} = U \Lambda U^\dagger$$

$$= \text{Tr} \left[\sqrt{U \Lambda U^\dagger \sigma U \Lambda U^\dagger} \right]$$

$$= \text{Tr} \left[\sqrt{U \Lambda \sigma \Lambda} U^\dagger \right] = \text{Tr} \left[U \sqrt{\Lambda \sigma \Lambda} U^\dagger \right]$$

$$= \text{Tr} \left[U \sqrt{\Lambda \sigma \Lambda} U^\dagger \right] \rightarrow \text{Apply cyclicity of trace}$$

$$= \text{Tr} \left[U^\dagger U \sqrt{\Lambda \sigma \Lambda} \right] = \text{Tr} \left[I \sqrt{\Lambda \sigma \Lambda} \right]$$

$$= \text{Tr} \left[\sqrt{\Lambda \sigma \Lambda} \right]$$

$$= \underline{\underline{F(\rho, \sigma)}}$$

2) Solution

$$[\rho, \sigma] = 0$$

$$\begin{aligned} \rho &= \sum_i p_i |i\rangle\langle i| \\ \sigma &= \sum_i s_i |i\rangle\langle i| \end{aligned} \quad \left. \begin{array}{l} \rho \text{ and } \sigma \text{ have same} \\ \text{eigenbasis} \end{array} \right\}$$

Proof:

$$F(\rho, \sigma) = F(\rho, \sigma)$$

$$F(\rho, \sigma) = \text{Tr} \left[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right] = \text{Tr} \left[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right]$$

$$= \text{Tr} \left[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right] = \text{Tr} \left[\sqrt{\sum_i p_i |i\rangle\langle i| \sum_j s_j |j\rangle\langle j| \sum_k p_k |k\rangle\langle k|} \right]$$

To make simple the step - first I will calculate the inside,

$$B \otimes B = \sqrt{\sum_i r_i |\psi_i\rangle\langle\psi_i|} \left(\sum_i s_i |\phi_i\rangle\langle\phi_i| \right) \sqrt{\sum_i r_i |\psi_i\rangle\langle\psi_i|}$$

For projector, has rank - 1: therefore

$$\sqrt{|\psi\rangle\langle\psi|} = |\psi\rangle\langle\psi|$$

$$\begin{aligned} B \otimes B &= \sqrt{\sum_i r_i |\psi_i\rangle\langle\psi_i|} \left(\sum_i s_i |\phi_i\rangle\langle\phi_i| \right) \sqrt{\sum_i r_i |\psi_i\rangle\langle\psi_i|} \\ &= \left(\sqrt{\sum_i r_i} \right)^2 \left(\sum_i s_i \right) \underbrace{|\psi_i\rangle\langle\psi_i|}_{=1} \underbrace{|\phi_i\rangle\langle\phi_i|}_{=1} \end{aligned}$$

$$= \sum_i r_i \sum_i s_i |\psi_i\rangle\langle\psi_i| = \sum_i r_i s_i |\psi_i\rangle\langle\psi_i|$$

Substitute in the main equation

$$F(\rho, \sigma) = \text{Tr} \left[\sqrt{\sum_i r_i s_i |\psi_i\rangle\langle\psi_i|} \right]$$

$$= \text{Tr} \left[\sum_i \sqrt{r_i s_i} |\psi_i\rangle\langle\psi_i| \right] = \sum_i \sqrt{r_i s_i} \text{Tr} [|\psi_i\rangle\langle\psi_i|]$$

$$= \sum_i \sqrt{r_i s_i} \cdot 1$$

$$= \sum_i \sqrt{r_i s_i}$$

$$F(\rho, \sigma) = F(r_i, s_i)$$

$$\underline{\underline{= 1/2}}$$

3) solution, the fidelity for pure state

The pure element for ρ and σ : $E = |\psi\rangle\langle\psi|$

$$\text{Pr}(\text{pass}) = \text{Tr}[E \cdot S]$$

$$= \text{Tr}[|\psi\rangle\langle\psi| B]$$

$$= \text{Tr}[B |\psi\rangle\langle\psi|]$$

6/2

$$\text{Tr}(\rho_{\text{pass}}) = \text{Tr} [\rho | \psi \rangle \langle \psi |]$$

$$= \underline{\underline{\langle \psi | \rho | \psi \rangle}}$$

$$\boxed{\text{for } \rho = |\psi\rangle\langle\psi|}$$

$$F(\psi, \rho) = \text{Tr} \left[\sqrt{\sqrt{\rho} \rho \sqrt{\rho}} \right]$$

$$\text{where, } \sqrt{\rho} \rho \sqrt{\rho} = \sqrt{|\psi\rangle\langle\psi|} \rho \sqrt{|\psi\rangle\langle\psi|}$$

$$= |\psi\rangle\langle\psi| \rho |\psi\rangle\langle\psi|$$

$$= \underline{\underline{(\langle \psi | \rho | \psi \rangle) |\psi\rangle\langle\psi|}}$$

$$F(\psi, \rho) = \text{Tr} \left[\sqrt{(\langle \psi | \rho | \psi \rangle) |\psi\rangle\langle\psi|} \right] = \text{Tr} \left[\sqrt{\langle \psi | \rho | \psi \rangle} |\psi\rangle\langle\psi| \right]$$

$$= \left(\sqrt{\langle \psi | \rho | \psi \rangle} \right) \text{Tr} [|\psi\rangle\langle\psi|] = \sqrt{\langle \psi | \rho | \psi \rangle} \cdot 1$$

$$= \sqrt{\langle \psi | \rho | \psi \rangle}$$

$$F^2(\psi, \rho) = \langle \psi | \rho | \psi \rangle$$

$$F^2(\psi, \rho) = \langle \psi | \rho | \psi \rangle$$

$$\underline{\underline{F^2(\psi, \rho) = \text{Tr}(\rho_{\text{pass}})}}$$

4) solution

$$F(|\psi\rangle, |\phi\rangle) = |\langle \psi | \phi \rangle|$$

$$F(|\psi\rangle, |\phi\rangle) = \text{Tr} \left[\sqrt{\sqrt{|\psi\rangle\langle\psi|} |\phi\rangle\langle\phi| \sqrt{|\psi\rangle\langle\psi|}} \right]$$

let's calculate first, the inside.

$$\begin{aligned} \overline{|1\psi\rangle\langle\psi|} |1\phi\rangle\langle\phi| \overline{|1\psi\rangle\langle\psi|} &= |1\psi\rangle\langle\psi| |1\phi\rangle\langle\phi| |1\psi\rangle\langle\psi| \\ &= |1\psi\rangle\langle\psi| |1\phi\rangle\langle\phi| |1\psi\rangle\langle\psi| \\ &= (|1\phi\rangle\langle\phi|) \cdot |1\psi\rangle\langle\psi| \end{aligned}$$

Sum of squares

$$\begin{aligned} F(|1\psi\rangle, |1\phi\rangle) &= \text{Tr} \left[\sqrt{|1\phi\rangle\langle\phi|^2 |1\psi\rangle\langle\psi|} \right] \\ &= \text{Tr} \left[\sqrt{(|1\phi\rangle\langle\phi|^2)} |1\psi\rangle\langle\psi| \right] \\ &= \sqrt{|1\phi\rangle\langle\phi|^2} \text{Tr} [|1\psi\rangle\langle\psi|] \\ &= \sqrt{|1\phi\rangle\langle\phi|^2} \cdot 1 = \sqrt{|1\phi\rangle\langle\phi|^2} \\ &= \underline{\underline{|1\phi\rangle\langle\phi|}} \end{aligned}$$

5) Solution, $B = |1\psi\rangle\langle\psi|$

Weed Variational (Helstrom) expression of trace distance

$$D(\rho, \sigma) = \max_{0 \leq M \leq I} \text{Tr} [M(\rho - \sigma)]$$

Let's take $m=3$

$$\begin{aligned} D(\rho, \sigma) &= \text{Tr} [B(\rho - \sigma)] = \text{Tr} [B^2 - B\sigma] = \underbrace{\text{Tr} [B^2]}_{=1} - \text{Tr} (B\sigma) \\ &= 1 - \text{Tr} (B\sigma) \\ &= 1 - \text{Tr} [|1\psi\rangle\langle\psi|\sigma] \\ &= 1 - \text{Tr} [\sigma |1\psi\rangle\langle\psi|] - \text{By trace cyclicity} \end{aligned}$$

$$= 1 - \langle \psi | \sigma | \psi \rangle$$

$$\xrightarrow{= 1}$$

$$D(\psi, \sigma) = 1 - \langle \psi | \sigma | \psi \rangle$$

$$= 1 - F(|\psi\rangle, \sigma)^2$$

$$D(\psi, \sigma) = 1 - F(|\psi\rangle, \sigma)^2$$

This is the maximum trace distance from all over operators
measurements in form

$$1 - F(|\psi\rangle, \sigma)^2 \leq D(\psi, \sigma)$$

Problem 3: Dynamic measures

2) Solution $F(|\psi\rangle, \rho) = \sqrt{\langle \psi | \rho | \psi \rangle}$

The fidelity between a pure state $|\psi\rangle$ and arbitrary
mixed quantum state ρ because $\rho = |\psi\rangle\langle\psi|$ is pure

$$F(\rho, \rho) = F(|\psi\rangle, \rho)$$

$$F(\rho, \rho) = F(|\psi\rangle, \rho) = \sqrt{\langle \psi | \rho | \psi \rangle}$$

$$= \sqrt{\langle \psi | \left((1-p) |\psi\rangle\langle\psi| + \frac{p}{2} \right) | \psi \rangle}$$

$$= \sqrt{\langle \psi | (1-p) |\psi\rangle\langle\psi| + \frac{p}{2} | \psi \rangle}$$

$$= \sqrt{(1-p) \cdot \langle \psi | \psi \rangle \langle \psi | \psi \rangle + \frac{p}{2} \langle \psi | \psi \rangle}$$

$$= \sqrt{(1-p) + \frac{p}{2}}$$

$$= \sqrt{2 - 2 + \frac{P}{2}}$$

$$= \sqrt{2 - \frac{2P + P}{2}}$$

$$= \sqrt{2 - \frac{P}{2}}$$

$$\underline{\underline{\quad}}$$

2) G-Power

$$\xi(\beta) = (4-1)\beta + P\alpha\beta\alpha$$

Required

$$D(\beta, \xi(\beta)) = ?$$

$$F(\beta, \xi(\beta)) = ?$$

$$D_{\min} = ?$$

$$F_{\min} = ?$$

Solution

$$D(\beta, \xi(\beta)) = \frac{1}{2} \text{Tr}[\beta - \xi(\beta)] = \frac{1}{2} \|\beta - \xi(\beta)\|$$

$$= \frac{1}{2} \|\beta - \xi(\beta)\|$$

$$= \frac{1}{2} \|\beta - (4-1)\beta - P\alpha\beta\alpha\|$$

$$= \frac{1}{2} \|\beta - \beta + \beta - P\alpha\beta\alpha\|$$

$$= \frac{1}{2} \|\beta - P\alpha\beta\alpha\|$$

Apply trace Norm on $\|\beta - P\alpha\beta\alpha\| = \|\beta(\beta - \alpha\alpha)\|$

$$\|\beta(\beta - \alpha\alpha)\| = 2\sqrt{2 - (4\psi|\alpha|\psi)^2} \text{ where } \beta = |\psi\rangle\langle\psi|$$

$$D(\beta, \xi(\beta)) = \frac{1}{2} (2\sqrt{2 - (4\psi|\alpha|\psi)^2})$$

$$D(\beta, \xi(\beta)) = \sqrt{2 - (4\psi|\alpha|\psi)^2}$$

$$D(S, \mathcal{E}(S)) = \sqrt{2 - (4\psi|\chi|\psi)^2}$$

$$F(S, \mathcal{E}(S)) = \text{Tr}[\sqrt{\mathcal{E}(S)} \mathcal{E}(S) \sqrt{\mathcal{E}(S)}] = \sqrt{4\psi|\mathcal{E}(S)|\psi} \quad \text{for pure state}$$

$$F(S, \mathcal{E}(S)) = \sqrt{4\psi|\mathcal{E}(S)|\psi}$$

pt calculate the inside

$$\begin{aligned} 4\psi|\mathcal{E}(S)|\psi &= 4\psi|(\alpha-\beta)S + \rho XBX|\psi \\ &= (\alpha-\beta) 4\psi|S|\psi + \rho 4\psi|XBX|\psi \\ &= (\alpha-\beta) 4\psi|\psi\rangle 2\psi|\psi\rangle + \rho 4\psi|X|\psi\rangle 2\psi|X|\psi\rangle \\ &= (\alpha-\beta) \cdot 2 \cdot 2 + \rho \underbrace{4\psi|X|\psi\rangle 2\psi|X|\psi\rangle}_{= (4\psi|X|\psi)^2} \\ &= (\alpha-\beta) + \rho (4\psi|X|\psi)^2 \end{aligned}$$

$$D(S, \mathcal{E}(S)) = \sqrt{(\alpha-\beta) + \rho (4\psi|X|\psi)^2}$$

The Dmin and Fmin

Dmin is at the specific value of $(4\psi|X|\psi) = 4\psi|X|\psi$

$4\psi|X|\psi$, has eigenvalue of ± 1

$$4\psi|X|\psi = \pm 1$$

$$(4\psi|X|\psi)^2 = (\pm 1)^2 = 1 //$$

$$D_{\min} = \rho(\sqrt{2-p})$$

$$= \rho(0)$$

$$= \underline{\underline{0}}$$

Fmin for pure state is at $|x\rangle = 0$

where $|x\rangle = 0$

$$F_{\min} = \sqrt{2-p + p \cdot 0}$$

$$= \sqrt{2-p + 0}$$

$$= \underline{\underline{\sqrt{2-p}}}$$

$$24|x+7| = 0$$

$$(24|x+7|)^2 = 0$$

$$\underline{\underline{F_{\min} = \sqrt{2-p}, \quad D_{\min} = 0}}$$

3) solution

$$F(|\psi\rangle, \mathcal{E}(\beta)) = F(|x\rangle, \mathcal{E}(\beta))$$

$$F(|x\rangle, \mathcal{E}(\beta)) = \text{Tr}[\sqrt{|x\rangle\langle x| \mathcal{E}(\beta) |x\rangle\langle x|}]$$

so this pure state I can use the following short expression

$$F(|x\rangle, \mathcal{E}(\beta)) = \sqrt{\langle x|x^\dagger (\mathcal{E}(\beta)) |x\rangle}$$

calculate the inside first

$$\langle x|x^\dagger (\mathcal{E}(\beta)) |x\rangle = \langle x|x^\dagger ((2-p)X^2 + pZ^2) |x\rangle$$

$$= (2-p) \langle x|x^\dagger X^2 |x\rangle + p \langle x|x^\dagger Z^2 |x\rangle$$

$$= (2-p) \langle x|X^2 |x\rangle + p \langle x|Z^2 |x\rangle$$

$$= (\lambda - p) \langle \psi | \beta | \psi \rangle + p \langle \psi | \alpha^2 | \psi \rangle \langle \psi | 2\alpha | \psi \rangle$$

$$= (\lambda - p) \langle \psi | 2\psi | \psi \rangle + p \langle \psi | \alpha^2 | \psi \rangle \langle \psi | 2\alpha | \psi \rangle$$

$$= (\lambda - p) \cdot 2 \cdot 1 + p \langle \psi | \alpha^2 | \psi \rangle \langle \psi | 2\alpha | \psi \rangle$$

Where:

$$\implies \langle \psi | \alpha^2 | \psi \rangle = (\langle \psi | 2\alpha | \psi \rangle)^*$$

Therefore $a \cdot a^* = |a|^2$

$$\langle \psi | x + (\lambda \beta) | \psi \rangle = (\lambda - p) + p |\langle \psi | \alpha^2 | \psi \rangle|^2$$

Substitute

$$F(x|\psi, \lambda(\beta)) = \sqrt{(\lambda - p) + p |\langle \psi | \alpha^2 | \psi \rangle|^2}$$

$$F(x|\psi, \lambda(\beta)) = \sqrt{(\lambda - p) + p |\langle \psi | \alpha^2 | \psi \rangle|^2}$$
